



UNIVERSITY OF
OXFORD

Towards Reliable AI in an Agentic Era

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University of Oxford

ChatGPT

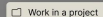
Health

 Claude Cowork

Let's knock something off your

Cowork is in research preview. Learn how to use it safely.

How can I help you today?

 Work in a project

 Pick a task, any task

 Optimize my week

Create a new project

Projects help Claude remember how you work. Files and instructions are stored in a folder on your computer.



Start from scratch

Set up a new folder with instructions and files.



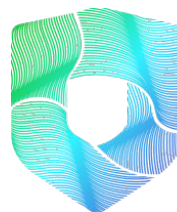
Import from chat

Bring over instructions and files from a chat project.



Use an existing folder

Give Claude a folder you already work from.



Hippocratic AI

— Do No Harm —

Deepseek

March 25, 2026

list the computer science faculty at pepperdine university



Here is the list of computer science faculty at Pepperdine University, based on the **Natural Science Division** (as Computer Science is part of that division, not a separate school).

Pepperdine's computer science program is relatively small and undergraduate-focused. The core faculty typically includes:

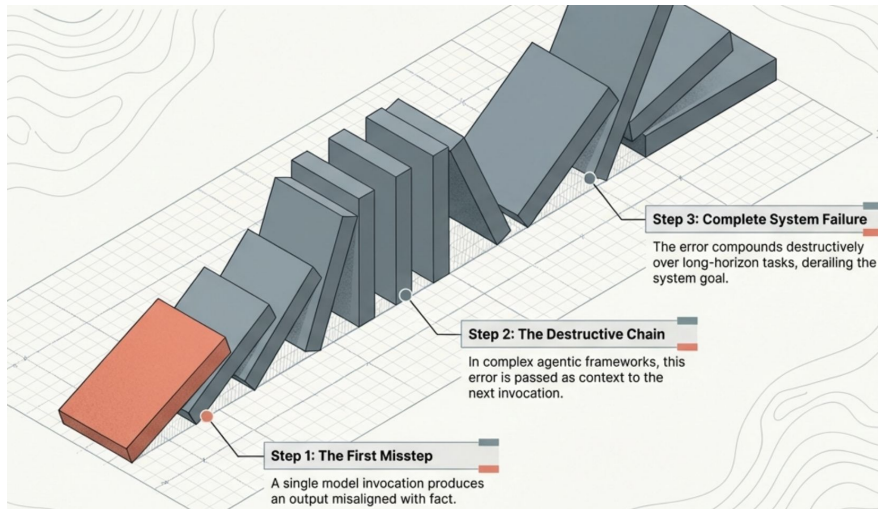
Core Faculty

- **Dr. Eric H. Johnson** (Professor of Computer Science)
 - *Area:* Algorithms, theoretical computer science, and computer science education.
- **Dr. Stephanie M. White** (Professor of Computer Science)
 - *Area:* Software engineering, formal methods, and requirements engineering.
- **Dr. Brian R. Spitzley** (Assistant Professor of Computer Science)
 - *Area:* Cybersecurity, human-computer interaction, and privacy.
- **Dr. Teomara (Teya) Rutherford** (Assistant Professor of Computer Science - joint appointment)
 - *Area:* Educational technology, games for learning, and data science.

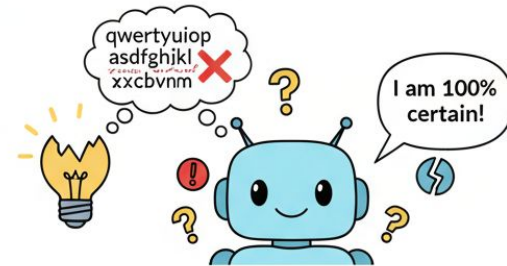
What are hallucinations and why do they matter?

Definition: when the model generates information that sounds plausible but is actually false, misleading, or unsupported by real data.

- Can mislead professionals in critical settings



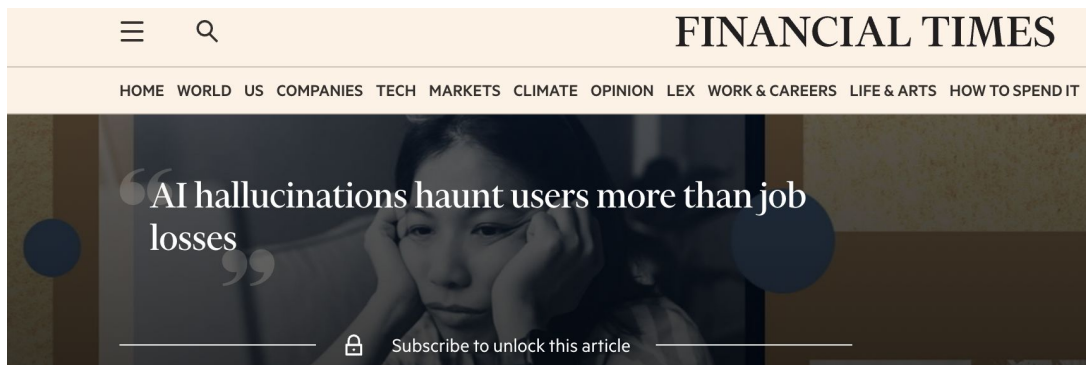
Risk of Hallucinations



CONFIDENTLY FALSE

Compounding Error in AI Agents

MIT report: 95% of generative AI pilots at companies are failing



"Chatbots still struggle with simple math or counting, depending on how you ask"

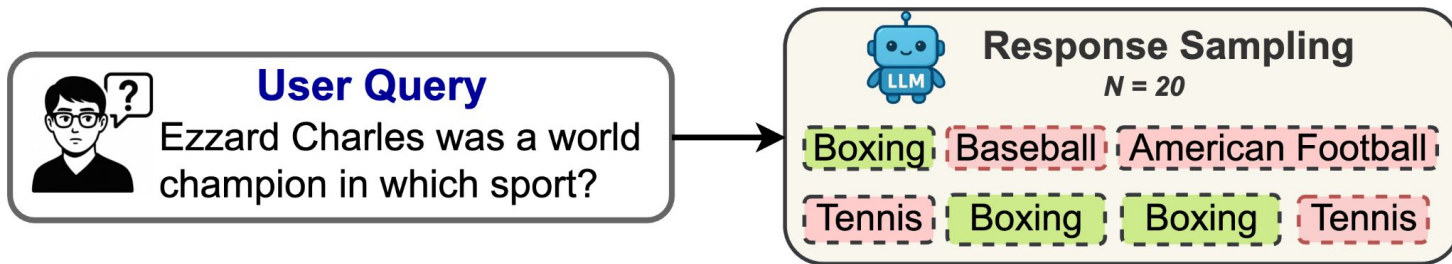
Geometric Uncertainty for Detecting and Correcting Hallucinations in LLMs



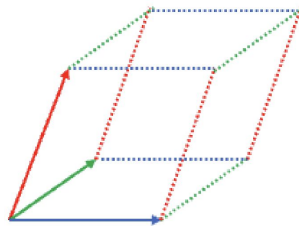
Under Review at TMLR

Sampling Based Approach

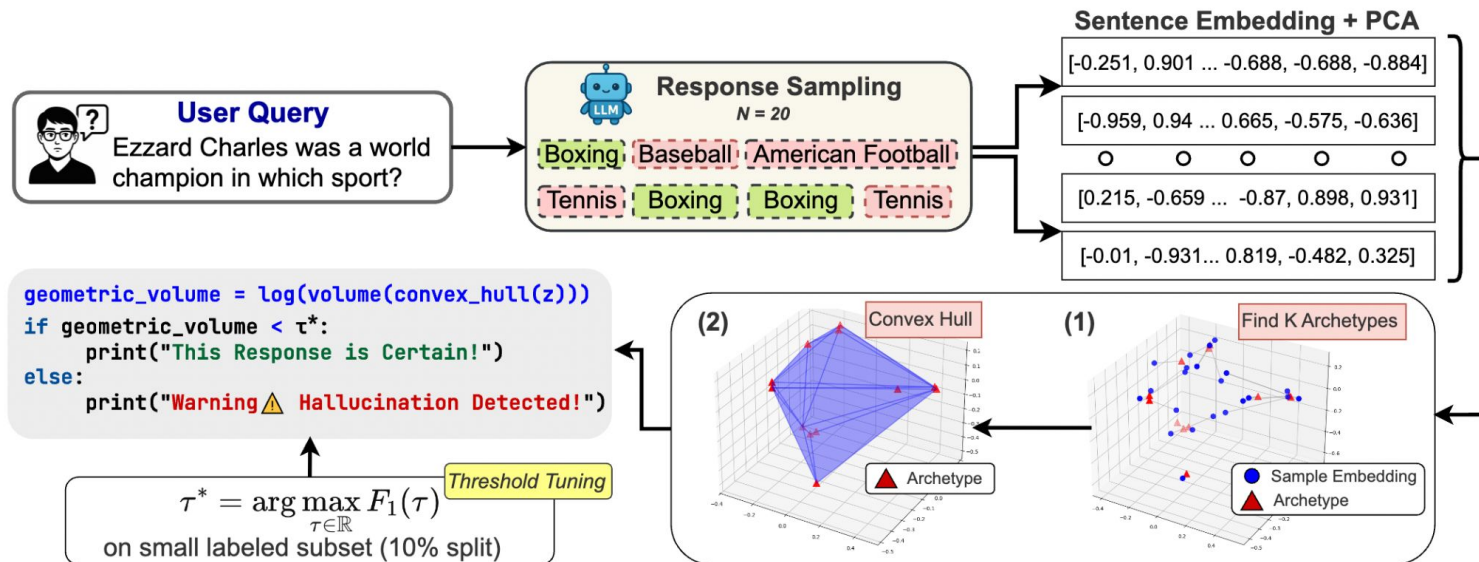
(1) Query an LLM the same question multiple times



(2) Measure how “different” responses are



Geometric Uncertainty



High “volume” = high chance of hallucination
Low “volume” = low uncertainty

Drawbacks of Sampling Based Approaches

- (1) Requires generating many responses → slow and expensive
 - *Can take 30 samples for a good estimate*
- (2) Not practical in real-time systems

Time complexity $O(M^2)$

- Not time sensitive
- Not cost sensitive

[nature](#) > [articles](#) > [article](#)

Article | [Open access](#) | Published: 19 June 2024

Detecting hallucinations in large language models using semantic entropy

[Sebastian Farquhar](#) , [Jannik Kossen](#), [Lorenz Kuhn](#) & [Yarin Gal](#)

[Nature](#) **630**, 625–630 (2024) | [Cite this article](#)

338k Accesses | **611** Citations | **1635** Altmetric | [Metrics](#)

Abstract

Large language model (LLM) systems, such as ChatGPT¹ or Gemini², can show impressive reasoning and question-answering capabilities but often ‘hallucinate’ false outputs and unsubstantiated answers^{3,4}. Answering unreliably or without the necessary information prevents adoption in diverse fields, with problems including fabrication of legal precedents⁵ or untrue facts in news articles⁶ and even posing a risk to human life in medical domains such

Semantic Self-Distillation for Language Model Uncertainty

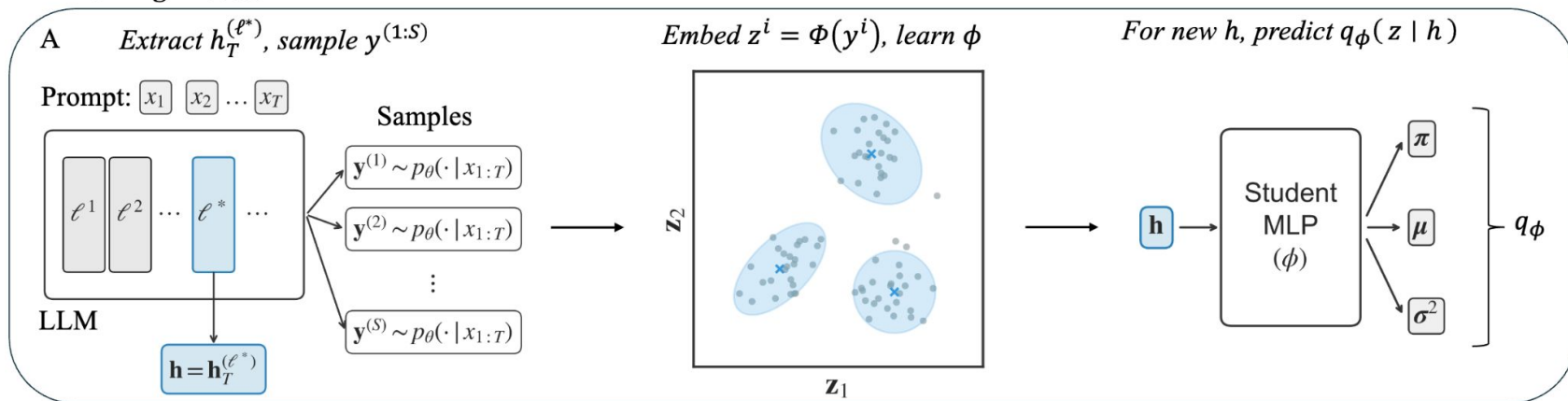


Under Review at UAI'26

Cheaper detection

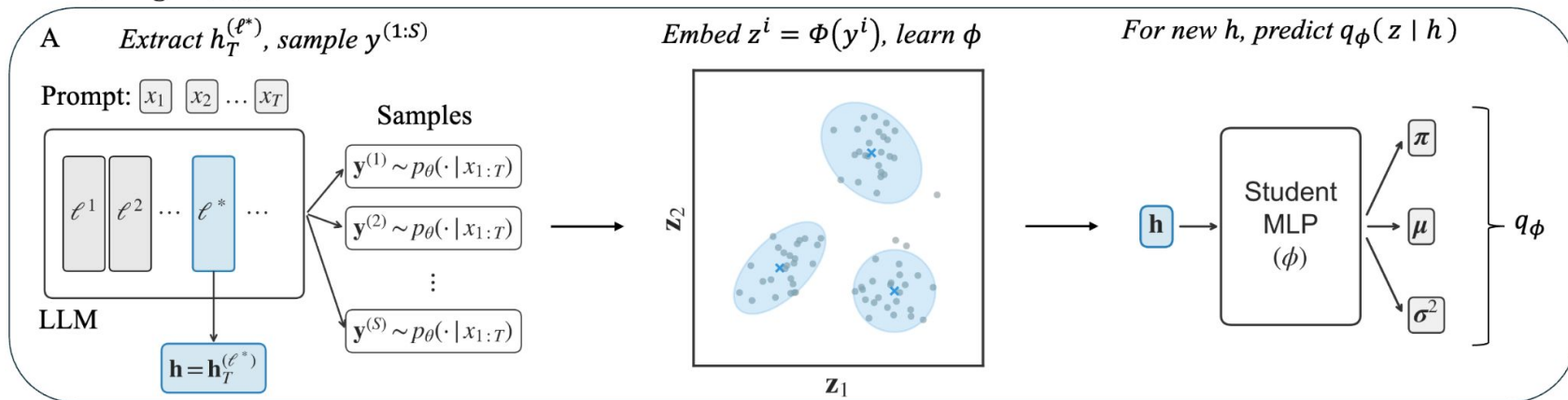
Instead of repeatedly sampling answers, can we predict that uncertainty in one shot?"

Training Process



Semantic Self Distillation

Training Process



Core Idea: Train a small model (MLP) to predict the uncertainty behavior of a large model (teacher)

Semantic Self Distillation

During Training:

1. Prompt: What is the capital of France?
 - Sample many answers: $y_1, y_2, \dots, y_n$
 - Eg. **Paris**, **Liege**, **Paris**, **Vienna**, **Paris**, **Paris**
 - Convert each answer $\rightarrow$ embedding z
 -
2. Prompt: What is the capital of France?
 - Pass prompt to LLM
 - Extract hidden state h (from middle layer)
3. Learn mapping from $h \rightarrow$ distribution over z with MLP

Sentence Embedding + PCA

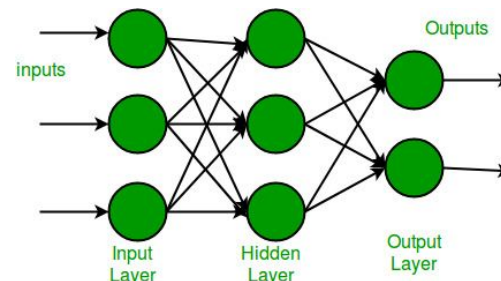
[-0.251, 0.901 ... -0.688, -0.688, -0.884]

[-0.959, 0.94 ... 0.665, -0.575, -0.636]

○ ○ ○ ○ ○

[0.215, -0.659 ... -0.87, 0.898, 0.931]

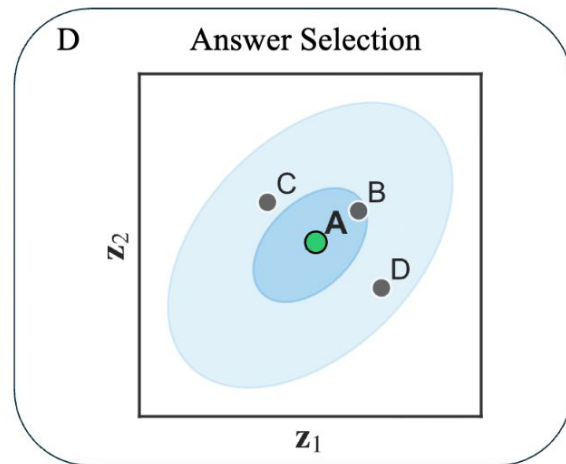
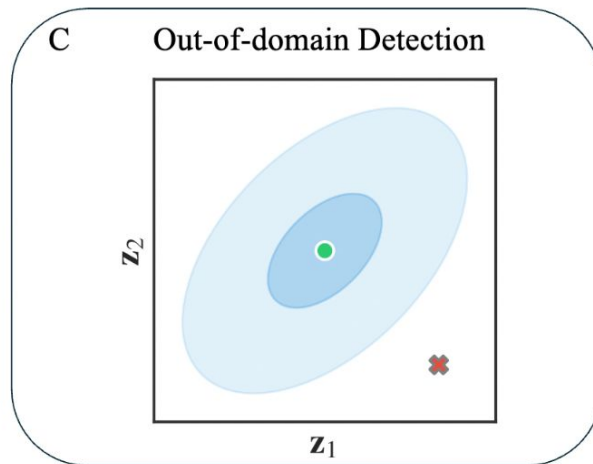
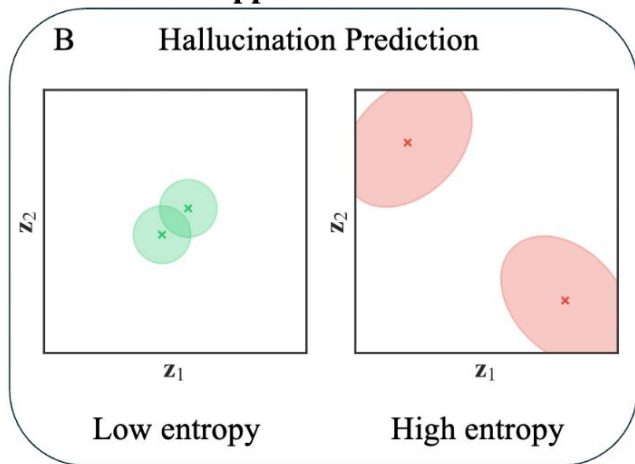
[-0.01, -0.931... 0.819, -0.482, 0.325]



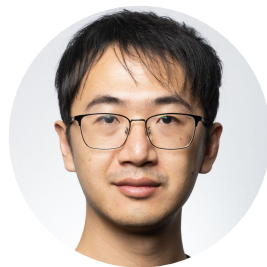
At inference time, we don't generate multiple answers anymore.

We just compute one vector h (from forward pass) and instantly predict the entire uncertainty distribution.

Inference Applications



BAS : A Decision-Theoretic Approach to Evaluating Large Language Model Confidence



Under Review at COLM'26

Measuring LLM Alignment

